(PV-TP-01) SwTP

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Product: `Portview`
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Document Overview

This document defines the integration-level software test procedures for Portview.

Document notes:

- The current procedure grouping reflects the integration-oriented workflow visible in the existing Portview test procedure content.
- The integration procedure is treated as `SwTP for Portview`, while the integration result record is treated as `SwTR-Z01 for Portview` in the current authored set.

Document Approval

Prepared by

Title	Name	Signature
Manager	`J. W. Lee`	

Reviewed by

Title	Name	Signature
General Manager	`S. I. Choi`	

Approved by

Title	Name	Signature
CTO (Director)	`K. Y. Ro`	

Revision History

Rev.	Date	Description
`0.0`	`2024-01-22`	Initial procedure set prepared according to verification-item subdivision
`0.1`	`2025-07-24`	Added translation test plan

1. Purpose

This document defines the integration test procedures used to verify Portview behavior across interacting software functions and interfaces.

2. References

This procedure set is based primarily on:

- `RS for Portview`
- `SV-04` Software Verification Plan

3. Test Information

Field	Value
Tester name	`J. W. Lee`
Tests performed date	`2025.07.24`
System location	`Genoray testing platform`
Configuration	`Default`
Software baseline	`Portview 2.2.5.16`

4. Procedure Coverage

The current integration procedure set visibly covers the following items.

Authored Procedure ID	Coverage
`ITP-001`	System instability check
`ITP-002`	Patient registration
`ITP-003`	Modify patient information
`ITP-004`	Patient search
`ITP-005`	Delete patient information
`ITP-006`	Export patient information
`ITP-007`	Select device
`ITP-008`	Acquire intraoral sensor image
`ITP-009`	Acquire image from digital camera
`ITP-010`	Acquire image from file
`ITP-011`	Acquire image from TWAIN device
`ITP-012`	Show status of device
`ITP-013`	Main viewer
`ITP-014`	Thumbnail viewer
`ITP-015`	Compare viewer
`ITP-016`	Tooth map window
`ITP-017`	Pan behavior
`ITP-018`	Zoom behavior
`ITP-019`	Reset
`ITP-020`	Delete
`ITP-021`	Rotate

Authored Procedure ID	Coverage
`ITP-022`	Text
`ITP-023`	Arrow
`ITP-024`	Pencil
`ITP-025`	Angle
`ITP-026`	Length
`ITP-027`	Print image
`ITP-028`	Print to DICOM printer
`ITP-029`	CD burning
`ITP-030`	Send image to DICOM server
`ITP-031`	Multi-sensor selection
`ITP-032`	Intraoral sensor FMX layout
`ITP-033`	USB integrity
`ITP-034`	Local-network connection check
`ITP-035`	Activity logging
`ITP-036`	GUI translation

4.1 Integration Procedure Groups

The current integration procedure coverage can be organized into stable test groups.

Procedure Group	Included Procedures
Patient and data workflow	`ITP-002` to `ITP-006`
Device acquisition workflow	`ITP-007` to `ITP-012`, `ITP-031` to `ITP-034`
Viewer and annotation workflow	`ITP-013` to `ITP-026`

Procedure Group	Included Procedures
Output and exchange workflow	`ITP-027` to `ITP-030`
Stability and support workflow	`ITP-001`, `ITP-035`, `ITP-036`

4.2 Relationship To Requirements

The integration procedures are expected to verify requirement interactions that originate from `RS for Portview` and flow into the software requirement set.

The current integration set especially supports:

- end-to-end patient and image workflow
- device-to-viewer and import-to-viewer transitions
- viewer-to-output and viewer-to-communication transitions
- integrity, logging, and localization behavior across user workflow states

4.3 Detailed Procedure Summaries

Each item below captures the authored integration-procedure intent at issue level. The summaries focus on cross-module workflow behavior, exception handling, and end-to-end acceptance outcomes.

4.3.1 Stability And Patient Workflow

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-001`	Portview with representative device and patient workflow	Exercise main workflow steps, induce documented error conditions where applicable, review manual guidance and logs	Errors are detectable, diagnosable, and aligned with documented recovery guidance
`ITP-002`	Portview in patient-list context	Register new patient data and exercise exception cases	Valid patient data is registered successfully and invalid inputs raise the expected message
`ITP-003`	Portview with existing patient record	Modify patient information and exercise exception cases	Modified patient data is saved correctly and invalid operations return the expected error handling

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-004`	Portview with searchable patient data	Search by patient identifiers and exercise search exception case	Matching patient data is retrieved correctly and failure conditions are reported properly
`ITP-005`	Portview with deletable patient record	Select and delete patient while avoiding active-record conflicts	Patient record is removed correctly and exception handling preserves data integrity
`ITP-006`	Portview with patient image data	Export patient-associated information or image data and exercise export exception	Export succeeds to selected destination and export errors are reported with the expected message

4.3.2 Device Acquisition And Source Integration

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-007`	Portview with one or more supported devices	Select device from device UI	Correct acquisition window or device path becomes active
`ITP-008`	Portview with intraoral sensor and selected patient	Select device, open patient, acquire image	Acquired sensor image is associated with the selected patient and displayed
`ITP-009`	Portview with digital camera and selected patient	Select camera source and acquire image	Camera image is acquired and displayed under the active patient context
`ITP-010`	Portview with supported file types	Import image file into active workflow	Imported image is loaded into the integrated viewer workflow
`ITP-011`	Portview with TWAIN device	Invoke TWAIN import and acquire image	TWAIN image is acquired and integrated into the current workflow

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-012`	Portview with connectable device state changes	Observe status handling before and after connection changes	Device status is reflected correctly in the integrated workflow state
`ITP-031`	Portview with multiple sensor options	Switch among sensors during workflow	Selected sensor is used consistently by subsequent acquisition actions
`ITP-032`	Portview with FMX layout workflow	Use FMX layout during acquisition sequence	FMX-related image placement and layout workflow behaves correctly
`ITP-033`	Installation or delivery media in integrated environment	Perform USB integrity verification before dependent workflow	Integrity issues are detected before they affect downstream workflow
`ITP-034`	Portview with network-dependent device path	Exercise local-network connection used by the device workflow	Network dependency is available and integrated acquisition path remains functional

4.3.3 Viewer And Annotation Workflow

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-013`	Portview with patient image set	Open patient and review main viewer state	Main viewer displays correct patient image context
`ITP-014`	Portview with multiple patient images	Observe thumbnail area after patient selection	Thumbnail viewer remains synchronized with active image set
`ITP-015`	Portview with comparable images	Open comparison workflow and apply comparison options	Comparison view is shown correctly for selected images
`ITP-016`	Portview with tooth-map workflow	Navigate by tooth map and review image changes	Image selection follows tooth-map interaction correctly

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-017`	Portview with displayed image	Pan displayed image	Image moves correctly while preserving workflow state
`ITP-018`	Portview with displayed image	Zoom in and out during active workflow	Zoom behavior is correct and image remains usable
`ITP-019`	Portview with modified viewer state	Use reset action after viewer manipulations	Viewer state returns to expected baseline presentation
`ITP-020`	Portview with deletable image or annotation context	Use delete action in permitted workflow	Target object is deleted without corrupting surrounding workflow state
`ITP-021`	Portview with displayed image	Rotate or flip image	Rotation or flip is correctly reflected in viewer state
`ITP-022`	Portview with annotation-enabled image	Add text annotation	Text annotation is created, displayed, and retained correctly
`ITP-023`	Portview with annotation-enabled image	Add arrow annotation	Arrow annotation is created, displayed, and retained correctly
`ITP-024`	Portview with annotation-enabled image	Add pencil annotation	Pencil annotation is created, displayed, and retained correctly
`ITP-025`	Portview with annotation-enabled image	Add angle measurement	Angle measurement is created and displayed correctly
`ITP-026`	Portview with annotation-enabled image	Add length measurement	Length measurement is created and displayed correctly

4.3.4 Output And Exchange Workflow

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-027`	Portview with printable image	Execute standard print flow	Printed output matches selected image and no application error occurs
`ITP-028`	Portview with configured DICOM printer	Execute DICOM-print flow	DICOM-print request is issued correctly and printer path completes or reports failure cleanly
`ITP-029`	Portview with media-export destination	Execute CD or media-burning export	Output media workflow completes and selected data is exported correctly
`ITP-030`	Portview with reachable DICOM server	Send image through DICOM transmission path	Transmission completes or returns controlled failure handling without data mismatch

4.3.5 Logging And Localization

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`ITP-035`	Portview with representative user operations	Perform integrated workflow and review logs	Log records reflect the executed workflow consistently
`ITP-036`	Portview with enabled language resources	Change GUI language during active workflow	Localized GUI remains coherent across integrated workflow screens

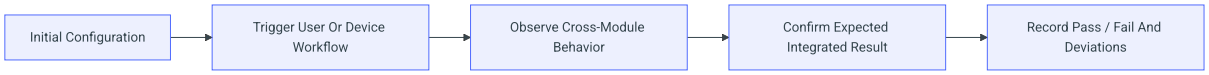
5. Integration Focus

The integration procedures are intended to demonstrate that:

- patient, study, and image data remain consistent across interacting software functions
- acquisition, display, annotation, and export flows operate together as intended
- device-state and communication functions behave correctly across workflow transitions

5.1 Procedure Execution Model

Each integration procedure should verify behavior across at least one multi-step workflow.



5.2 Evidence Expectations

Each executed integration procedure should produce or support:

- procedure execution date
- executor identification
- configuration under test
- observed integrated behavior
- pass or fail status
- deviation note where applicable

6. Current Test Environment Basis

The currently visible controlled procedure content identifies:

Environment Element	Current Basis
Test performer	`J. W. Lee`
Test date	`2025.07.24`
Test location	`Genoray testing platform`
Configuration	`Default`
Software under test	`Portview 2.2.5.16`

7. Relationship To The SV Set

This procedure set supports:

- integration planning in `SV-04`
- integration result summarization in `SV-05`
- traceability linkage between requirements and verification evidence